higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N5

26 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 $23,5_{10} – C,4_{16}$

$23,5_{10}$

Laborious method (table method):

2^4	2^3	2^2	2^1	2^0	2^{-1}
1	0	1	1	1	1

$23,5_{10} = 10111,1_2$✓

$C,4_{16} = 1100,0100_2 = 1100,01_2$✓ (each hexadecimal is represented by 4 binary bits)

10111,01 ⟶ 1 0 1 1 1,1 0
-01100,01 1sc ⟶ 10011,10 + 1 2sc ⟶ 1 0 0 1 1,1 1✓
There is a carry, the answer will be positive 1 0 1 0 1 1,0 1✓

Ignore the carry 01011,01✓

For hexadecimal group binary in groups of 4:

∴ $10111,1_2 – 1100,01_2 = 1011,0100_2 = B,4_{16}$✓ (6)

1.2 $34,25_{10} \times 107,5_8$

$34,25_{10}$

Laborious method (table method):

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
1	0	0	0	1	0	0	1

$34,25_{10} = 1000101,01_2$√

$107,5_8 = 001\ 000\ 111_2 = 1000111,101_2$√

$34,25_{10} \times 107,5_{10} = 1000101,01_2 \times 1000111,101_2$

```
                  1 0 0 0 1 0 0 1
              1 0 0 0 1 1 1 1 0 1
              ___________________
                  1 0 0 0 1 0 0 1 √
                1 0 0 0 1 0 0 1 0 0
              1 0 0 0 1 0 0 1 0 0 0
            1 0 0 0 1 0 0 1 0 0 0 0√
          1 0 0 0 1 0 0 1 0 0 0 0 0
    1 0 0 0 1 0 0 1 0 0 0 0 0 0 0 0
    ___________________________________
  1 0 0 1 1 0 0 1 0 1 0 1, 0 0 1 0 1√
```

Therefore $1\ 0\ 0\ 1\ 1\ 0\ 0\ 1\ 0\ 1\ 0\ 1, 0\ 0\ 1\ 0\ 1 = 2453,15625_{10}$√ (6)

1.3 $132,5_8 \div 21,A_{16}$

$132,5_8 = 001\ 011\ 010,101_2 = 1011010,101_2$ (each octal is represented by 3 binary bits)✓

$21,A_{16} = 0010\ 0001,1010_2 = 100001,101_2$ (each hexadecimal is represented by 4 binary bits)✓

```
                        1 0, 1 0 1✓
                    ___________________
100001101√ 1 0 1 1 0 1 0 1 0 1,  0 0 0
           100001101 0
           ___________
             1 0 1 1 1 0 1 1 0 0✓
             1 0 0 0 0 1 1 0 1 0
             ___________________
               1 1 0 1 0 0 1 0 0 0
               1 0 0 0 0 1 1 0 1 0✓
               ___________________
                 1 0 0 1 0 1 1 1 0
```

$\therefore\ 1011010,101_2 \div 100001,101_2 = 10,101_2$ remainder $\frac{100011010}{100101110}$

= 10,101 (for octal group binary in groups of 3)

$= 2,5_8$✓ (6)

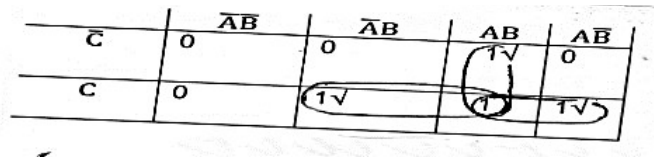
[18]

QUESTION 2

Truth table:

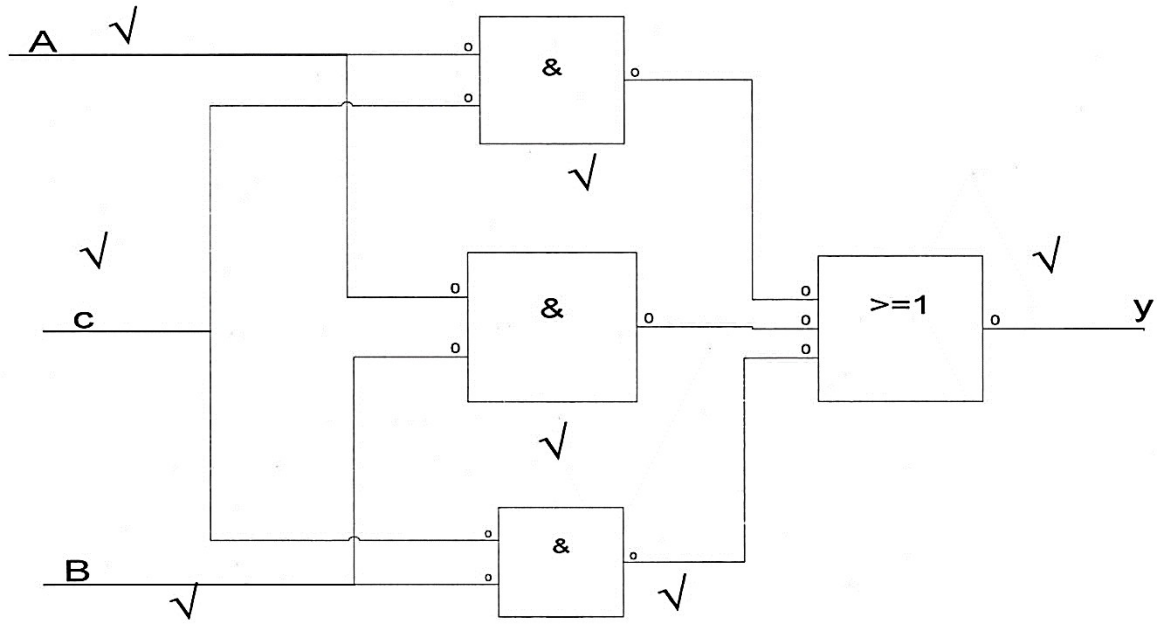
A	B	C		Y	
0	0	0		0	
0	0	1		0	✓
0	1	0		0	✓
0	1	1		1	✓
1	0	0		0	✓
1	0	1		1	✓
1	1	0		1	✓
1	1	1		1	✓

Karnaugh map:



Y = BC+ AC + AB✓✓

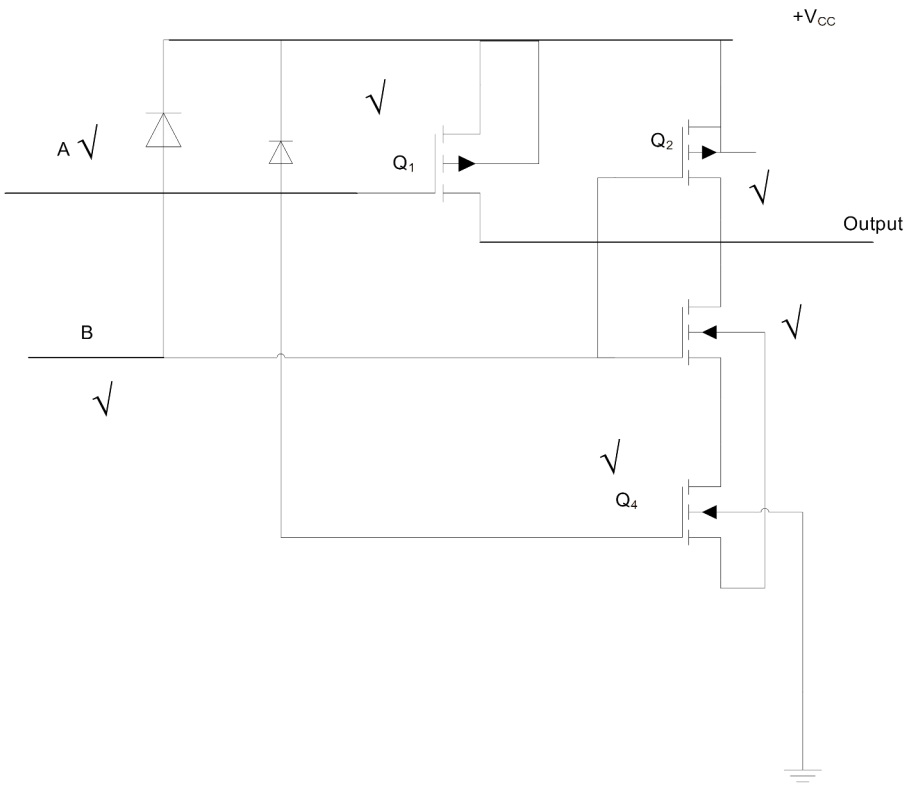
Circuit diagram:



[20]

QUESTION 3

3.1



(6)

3.2
3.2.1 True
3.2.2 True
3.2.3 True
3.2.4 False
3.2.5 False

(5 × 2) (10)
[16]

QUESTION 4

4.1
4.1.1 C
4.1.2 A
4.1.3 E
4.1.4 B
4.1.5 D

(5 × 1) (5)

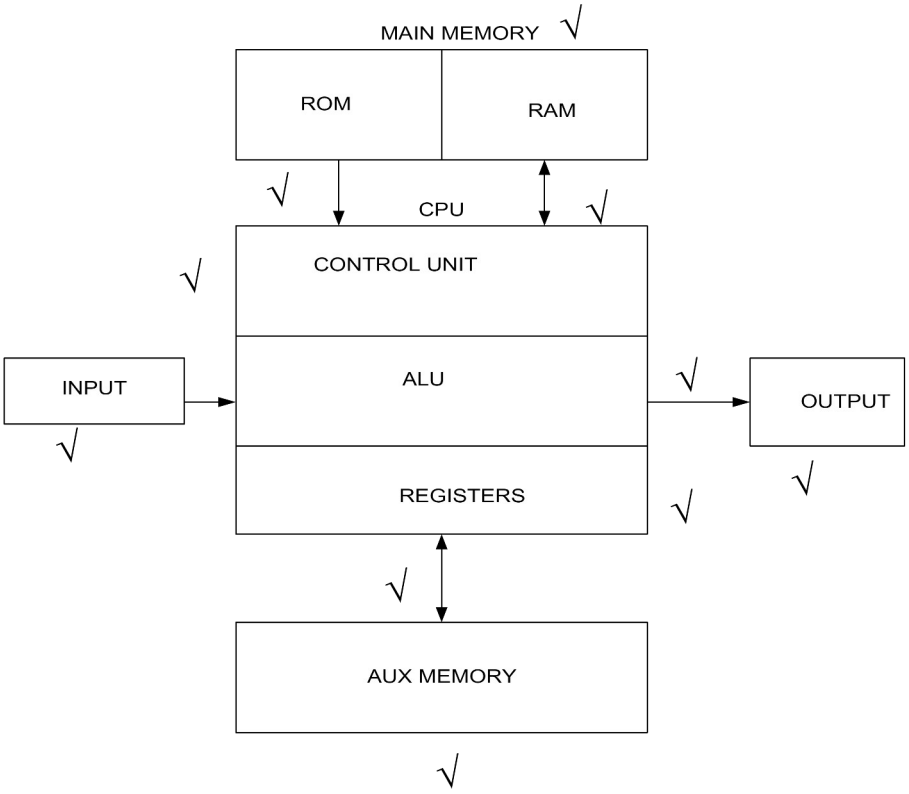
4.2
- Electrostatic
- Thermal
- Inkjet
- Xerographic
- Laser type
- Dot

(Any 4 × 1) (4)

4.3 UART converts parallel data received from the CPU to serial data for transmission over long distance and also converts serial data received to parallel data for processing by the CPU. (2)
[11]

QUESTION 5

5.1



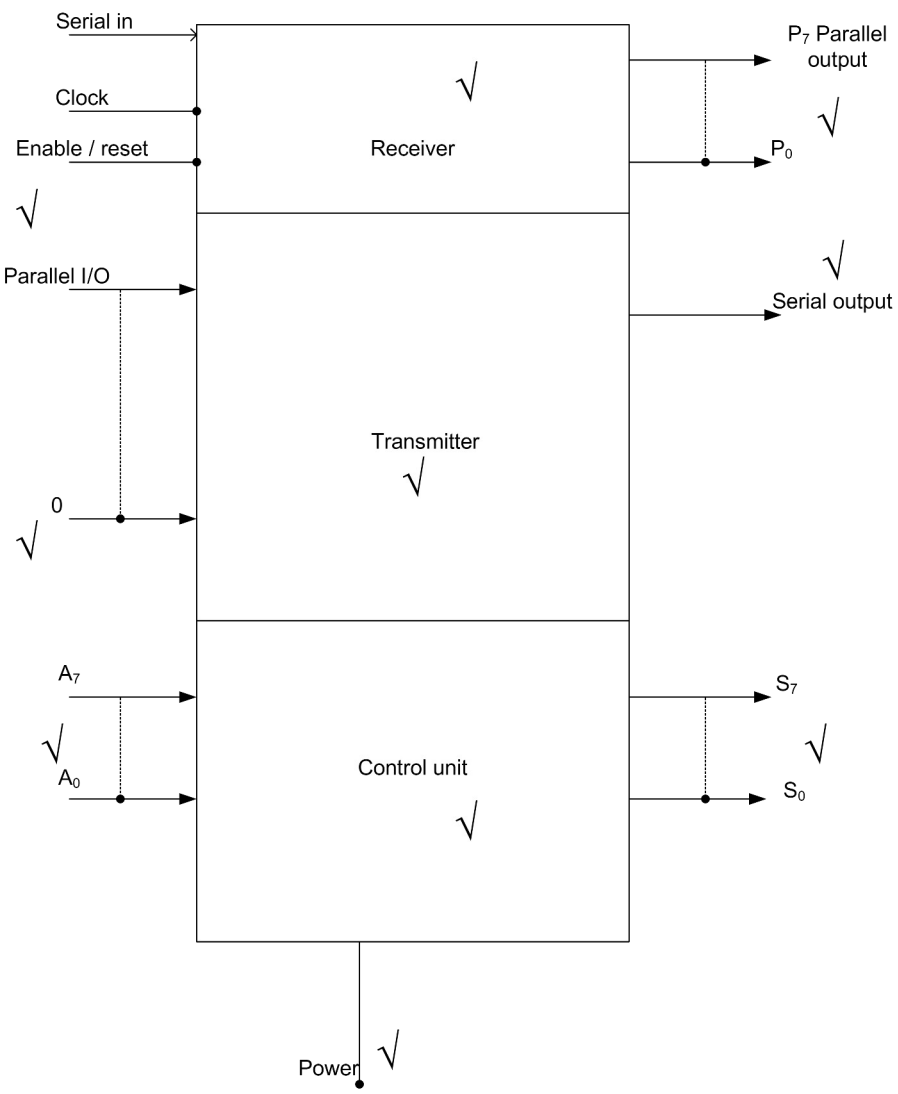
(10)

5.2
- Computer display
- Printer
- Projector
- Computer speaker
- Game controller (5)

[15]

QUESTION 6

6.1



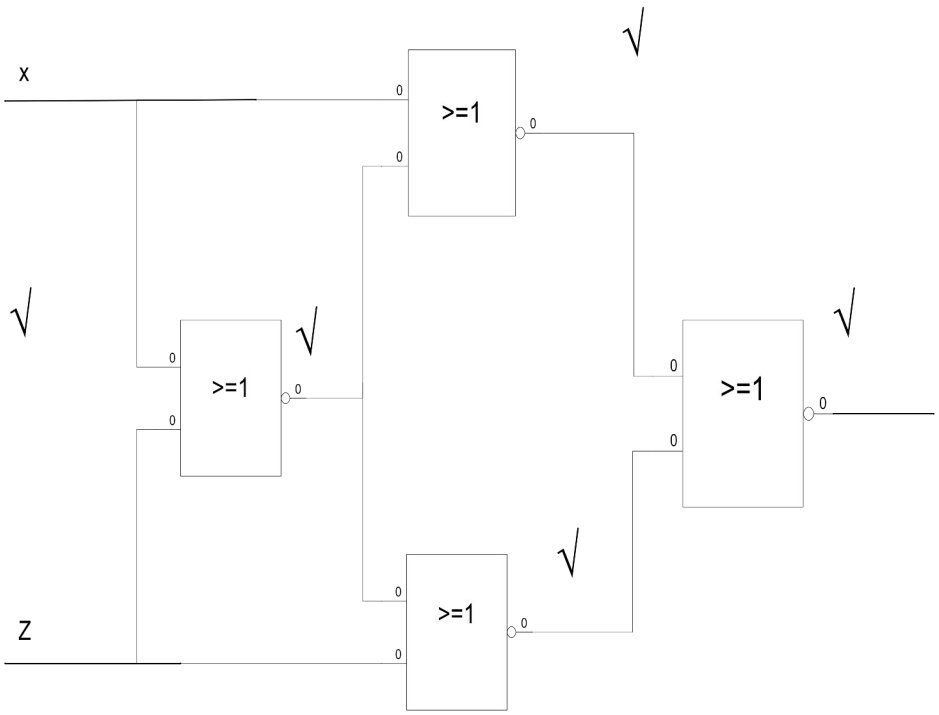
(9)

6.2 RAM has the disadvantage off being volatile, meaning when the supply or power is off, all data will be lost. (1)

[10]

QUESTION 7

7.1



(5)

7.2

7.2.1	D
7.2.2	B
7.2.3	D
7.2.4	C
7.2.5	D

(5 × 1) (5)
[10]

TOTAL: 100

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